Synthesis Fit Validation Study

Current scoped real-JD validation, production-fit interpretation, and proposed human check

Executive Summary

This study evaluates whether the Synthesis Fit Analyzer preserves a coarse occupational signal when scoring real resumes against real job postings. The current validation is scoped to three O*NET-aligned families: Consultant, Finance, and Information Technology. Each resume is scored against posting-level JDs from each family, and the method is credited when the true family receives the highest average score.

- The best pre-specified method is **Hybrid 0.25 rules / 0.75 semantic**, with top-1 accuracy of **68.0%**, MRR of **0.825**, and mean rank of **1.41**.

- This is strong enough for a graduate course validation study, but it should be interpreted as a coarse discriminative sanity check, not as direct pairwise fit accuracy.

- A post-hoc grid search over hybrid weights found a maximum family-level top-1 of 68.8 percent at structured weight 0.15. Because that was test-set tuning and did not cross the 70 percent threshold, the report keeps the pre-specified hybrid 0.25 as the main candidate.

- The next, more task-valid evaluation should manually label 40-60 JD-resume pairs as strong, medium, or weak fit and test whether analyzer scores rank them in that order.

Why This Validation Exists

The production Fit Analyzer is a one-JD-to-one-resume scorer. A family-level task cannot prove that a specific score is correct for a specific pair. Instead, this validation asks a narrower question: does a scoring method generally place a resume closer to postings from its own occupational family than to postings from other scoped families? This is useful as a large-scale sanity check before doing a smaller human-labelled pair-level validation.

Study Design

- Input data: local resume and posting datasets stored under the repo's gitignored Datasets directory.

- Family mapping: candidate postings are classified by the LLM mapper into 21 retained families plus UNMAPPED. The current validation then filters to Consultant, Finance, and Information Technology.

- JD sampling: 100 high-confidence postings are collected for each scoped family before scoring.

- Parser gate: before scoring, each selected JD is parsed with the same production parseJD() function. The main study keeps JDs with at least three parsed requirements.

- Scoring unit: each resume is scored against every retained JD. Scores are averaged by JD family, then the family with the highest average score is the predicted family.

- Same split for all methods: rules-only, embedding-only, and all hybrid arms use the same resumes and JDs.

Dataset After Parsing

Family	Resumes	JDs kept	JDs originally sampled
CONSULTANT	115	95	100
FINANCE	118	94	100
INFORMATION-TECHNOLOGY	120	80	100
Total	**353**	**269**	**300**

The parseability gate kept **269** of **300** postings. The threshold was at least **3** parsed requirements per JD.

Family	Kept	Dropped	Mean req.	Median req.	Zero req.
CONSULTANT	95/100	5	12.65	14.0	0
FINANCE	94/100	6	13.22	14.0	0
INFORMATION-TECHNOLOGY	80/100	20	8.96	7.5	3

Methods Compared

Method	Description	Production interpretation
Rules-only structured	Current deterministic scoreFit() logic: O*NET-grounded skill extraction, requirement status, gaps, and evidence.	Strong baseline and most interpretable path.
Embedding-only semantic	Requirement-level semantic retrieval over resume evidence chunks using local BGE-small embeddings.	Tests whether semantic similarity aligns better with human-like fit judgement.
Hybrid arms	Per-resume min-max blend of structured and embedding family scores with structured weights 0.25, 0.50, and 0.75.	Hybrid 0.25 is now the production candidate, subject to human check.

Headline Results

Arm	Top-1	Mean rank	MRR	Mean margin
Rules-only structured	54.7%	1.63	0.743	2.60
Embedding-only semantic	65.2%	1.43	0.813	0.27
Hybrid 0.25 rules / 0.75 semantic	**68.0%**	1.41	0.825	0.18
Hybrid 0.50 rules / 0.50 semantic	64.3%	1.47	0.803	0.18
Hybrid 0.75 rules / 0.25 semantic	56.1%	1.58	0.756	0.12

Top-3 accuracy is intentionally not reported as a headline metric because this validation has only three families. The more informative metrics are top-1, mean rank, MRR, margin, and the confusion matrix.

Per-Family Findings

Family	Structured	Embedding	Hybrid 0.25	Main observation
CONSULTANT	39.1%	34.8%	38.3%	Hardest family; business, finance, and consulting language overlaps heavily.
FINANCE	100.0%	78.0%	89.0%	Very strong across structured-heavy methods; financial requirements are distinctive.
INFORMATION-TECHNOLOGY	25.0%	81.7%	75.8%	Embedding improves sharply after parser robustness fixes.

Confusion Matrix - Best Arm

Rows are true resume families; columns are predicted families. Values are resume counts.

True / Predicted	Consultant	Finance	IT
Consultant	44	21	50
Finance	3	105	10
IT	22	7	91

Interpretation

- The hybrid 0.25 arm is the best pre-specified family-level proxy method. It balances semantic flexibility with some rules-based grounding.

- The 70 percent threshold is not formally met. However, 68.0 percent on 353 resumes is close enough that the result should be discussed as near-threshold rather than as a hard method failure.

- Consultant is the main failure mode. This is expected because consultant resumes and JDs often share general business, analysis, stakeholder, strategy, and finance vocabulary with the other scoped families.

- Family-level errors can be reasonable transferability cases. A consultant resume scoring highly for a finance analyst JD is not necessarily a bad fit in the production task.

Limitations

- **Proxy task:** family top-1 measures coarse occupational discrimination, not one-pair fit accuracy.

- **Label source:** posting family labels come from the LLM mapper and have not yet been human-checked.

- **Scope:** only three families are included, so results should not be generalized to all O*NET families.

- **Parser dependency:** the study depends on parseResume() and parseJD(); parser errors can affect scores.

- **Ambiguous fit:** cross-family transferability is treated as wrong by family top-1 even when it may be reasonable for a real applicant.

- **Weight tuning:** a post-hoc sensitivity check found 68.8 percent at structured weight 0.15, but this is not adopted as the main method because it would tune to the family proxy test set.

Proposed Human Check

The next validation should be a smaller but more task-valid pair-level study. Instead of asking whether a resume belongs to the same broad family as a JD, it should ask whether a specific resume is a strong, medium, or weak fit for a specific JD.

- Sample 40-60 JD-resume pairs, stratified by analyzer score: about one third high-score, one third middle-score, and one third low-score pairs.

- Include all three scoped JD families and a few plausible cross-family pairs, especially Consultant versus Finance.

- Hide analyzer scores during labelling to reduce confirmation bias.

- Label each pair with a fixed rubric, then compare rules-only, embedding-only, and hybrid 0.25 against those labels.

Human Labelling Rubric

Dimension	0	1	2
Core skills match	Most must-have skills missing	Some core skills matched	Most core skills matched
Experience and domain	Clearly unrelated	Transferable overlap	Highly related work/domain
Seniority and years	Clearly below or mismatched	Close but imperfect	Meets level and years
Education or hard constraints	Hard requirement missing	Unclear or partial	Meets or not required

Total score mapping: 0-3 = Weak fit, 4-6 = Medium fit, 7-8 = Strong fit. This makes human labels more objective than a single impressionistic judgement.

Human Check Metrics

- **Spearman correlation:** analyzer score versus human total score.

- **Mean score monotonicity:** Strong pairs should have higher average analyzer scores than Medium, which should be higher than Weak.

- **Pairwise ordering accuracy:** for two pairs with different human labels, the method should rank the stronger human label higher.

- **Optional threshold view:** inspect whether score bands can map to strong, medium, and weak recommendations.

Recommended Reporting Position

Use the current family-level validation as large-scale supporting evidence, not as the final claim of pairwise accuracy. The main claim should be: hybrid semantic-plus-rules scoring shows stronger coarse occupational signal than rules alone, with hybrid 0.25 as the best pre-specified validation arm. Final production calibration should be based on the proposed human-labelled JD-resume pair study.

Reproducibility: run npm run validate:prep, npm run validate:fit, and npm run validate:report. The report uses scripts/validation/.artifacts/metrics.scoped.json generated by that pipeline.